Changes from feedback survey:

(for relevant sections) move desk more gently coming into class

Walk around during worksheet review time

Review time remains 15m. 1/3, 1/3, 1/3 of students voted increase, decrease, no prefer.

(for relevant sections) Solve projector Zoom. Unfortunately, no solution. I called classroom tech 3 times, looked for remote, tried buttons on projector. Zoom is fixed

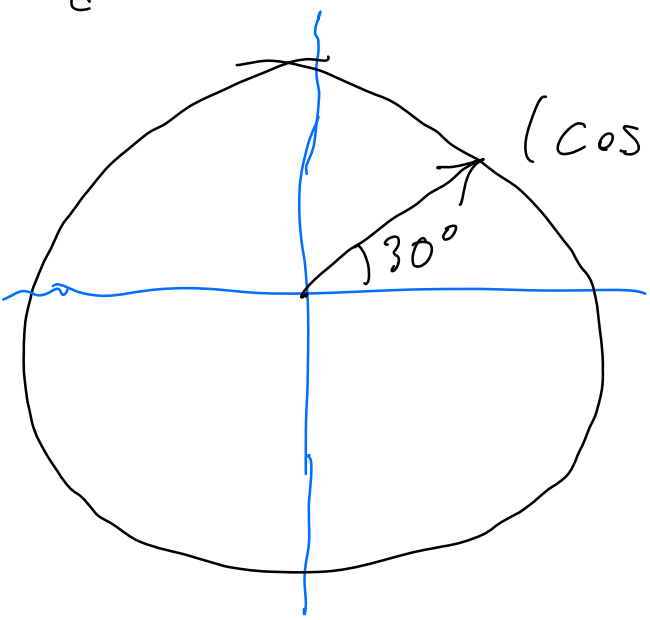
ASK about questions outside worksheet & don't forget to ask about previous worksheets, time permitting

Encourage collaboration. You are free to work on worksheet with other people, in groups, before class, after class, on a train, on a plane, eating green eggs

& 2am, etc.

16: $g = x^2y$ directional derivative at $(2, e)$ in direction angle $\frac{\pi}{6}$ above positive x-axis

Step 1: convert direction info into actual unit vector for directional derivative formula



$(\cos 30°, \sin 30°) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ is 30° above x-axis on unit circle, so

$\left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle$ is the direction.

Step 2: apply formula using g & direction

DD will be $\nabla g \cdot \hat{v}$ in direction of

a unit vector $\hat{v}$.

$g = x \ln y \Rightarrow \nabla g = \left\langle \ln y, \frac{x}{y} \right\rangle$, so

$\nabla g(2, e) = \left\langle 1, \frac{2}{e} \right\rangle$, so

$$\nabla g \cdot \hat{v} = \left\langle 1, \frac{2}{e} \right\rangle \cdot \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle$$

$$= \boxed{\frac{\sqrt{3}}{2} + \frac{1}{e}}.$$

4c: classify critical points of

$h = (x + y^2) e^x$

Step 1: Solve $\vec{0} = \nabla h$ for crit pts

$$\vec{0} = \left\langle (1 + x + y^2) e^x, \; 2y e^x \right\rangle$$

$0 = 2y e^x \Rightarrow y = 0$ because $e^x \neq 0$

$0 = e^x + x e^x = (x+1) e^x \Rightarrow x = -1$,

so only crit pt. is $(-1, 0)$

Step 2: Apply 2nd derivative test

$$H = \begin{bmatrix} (2+x+y^2)e^x & 2ye^x \\ 2ye^x & 2e^x \end{bmatrix}, \text{ so}$$

$$\det = f_{xx}f_{yy} - (f_{xy})^2 =$$

$$e^{2x}\left[2(2+x+y^2) - (2y)^2\right] =$$

$$e^{2x}\left[4 + 2x + 2y^2 - 4y^2\right] =$$

$$2e^{2x}\left[2 + x - y^2\right], \text{ so}$$

$$\det(-1,0) = 2e^{-2}\left[1 - 0^2\right] = 2e^{-2} > 0,$$

so $(-1,0)$ is a local minimum & no other extrema

5b: find absolute min/max of $f = x + y^2 + x^2y + 9$ on $D = \{|x|, |y| \le 1\}$

and $f = x^2+y^2-2x$, D triangle bounded by $(2,0), (0,2), (0,-2)$

Either part: solve $\nabla = 0$ for points inside region & check min/max on boundary, then compare the two

$\langle 0,0 \rangle = \nabla f = \langle 2x(1+y), 2y+x^2 \rangle$

$0 = 2x(1+y) \Rightarrow x=0$ or $y=-1$

$y = -\frac{1}{2}x^2$ or $x = \pm\sqrt{-2y}$

$x=0 \Rightarrow y = -\frac{1}{2}\cdot 0^2 = 0$

$y=-1 \Rightarrow x = \pm\sqrt{2}$, so

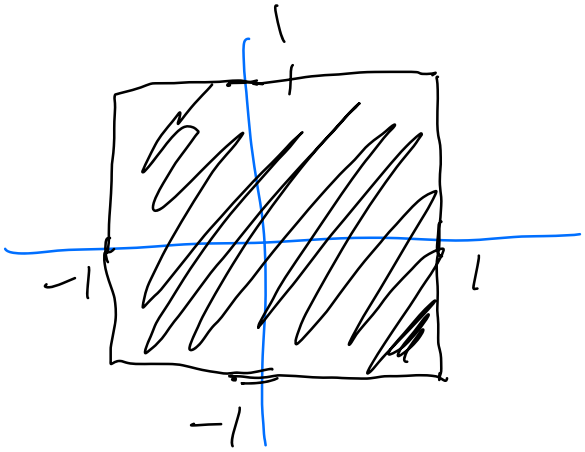
$(\pm\sqrt{2}, -1)$, $(0,0)$ are crit points

Now let's check these values

$f(0,0) = 9$

$f(\pm\sqrt{2}, -1) = 2 + 1 - 2 + 9 = 10$

Now let's check boundary



From graphing, region is a square & has boundary as its perimeter, which is $x = \pm 1$, $y = \pm 1$

$x = \pm 1$: $f = x^2(1+y) + y^2 + 9 =$

$y^2 + y + 10 = \left(y + \frac{1}{2}\right)^2 + 9.75 \geq$

9.75, so minimum is 9.75 & maximum is 12.

$y = 1$: $f = 2x^2 + 10$, which has min 10 & max 12

$y = -1$: $f = 10$

By comparing all values,

9 & 12 are absolute min & max

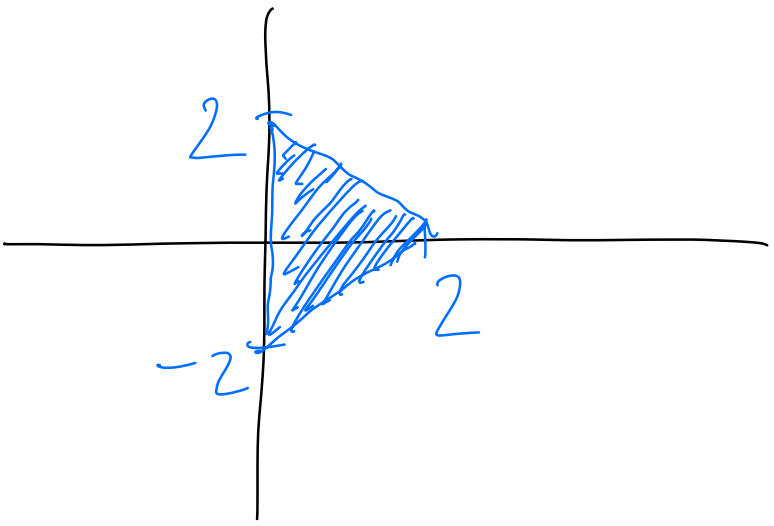
$f = x^2 + y^2 - 2x$

$\langle 0,0 \rangle = \nabla f = \langle 2x-2, 2y \rangle$, so

$2x-2 = 0 \Rightarrow x = 1$, $2y = 0 \Rightarrow y = 0$,

so $(1,0)$ is only critical point.

$f(1,0) = 1 - 2 = -1$



Boundary of region is 3 sides of triangle, which are

$x = 0$ $x - 2 \le y \le 2$, $y = x - 2$, $y = 2 - x \Rightarrow y = \pm(x-2)$, $0 \le x \le 2$

$y = \pm(x-2)$: $f = x^2 + (x-2)^2 - 2x =$

$2x^2 - 2x + 4 = 2(x^2 - x + 2) =$

$2\left[\left(x - \frac{1}{2}\right)^2 + 1.75\right) \geq 2 \cdot 1.75 =$

3.5 and max is at $x = 2 \Rightarrow$

$2(4) = 8$

$x = 0$: $f = y^2 \in [0, 4]$

By comparing all values,

-1 & 8 are min & max

6: profit is $f = 48x + 96y - x^2 - 2xy - 9y^2$,

$x =$ tennis balls (1000s), $y =$ hours of advertising, can produce ≤ 50000 balls per month, purchase ≤ 25 hr advertising, find x&y that maximize profit

Step 1: find constraints on x & y

can produce ≤ 50K balls & must produce ≥ 0, so $0 \leq x \leq 50$

can purchase ≤ 25 hr & must purchase ≥ 0, so $0 \leq y \leq 25$, so

$(x, y) \in [0, 50] \times [0, 25] = D$ &

we want to maximize f on D.

Step 2: solve $0 = \nabla f$ inside D.

$$\langle 0, 0 \rangle = \nabla f = \langle 48 - 2x - 2y, 96 - 2x - 18y \rangle$$

$$\Rightarrow \begin{cases} 24 = x + y \\ 48 = x + 9y \end{cases}$$

$24 = (x + 9y) - (x + y) = 8y \Rightarrow y = 3 \Rightarrow$

$x = 24 - y = 21$, so

$(21, 3)$ only crit point.

$f(21,3) = (48-21)\cdot 21 - 3\cdot(96-9\cdot 3) - 2\cdot 21\cdot 3 = 27\cdot 21 - 3\cdot 69 - 126$

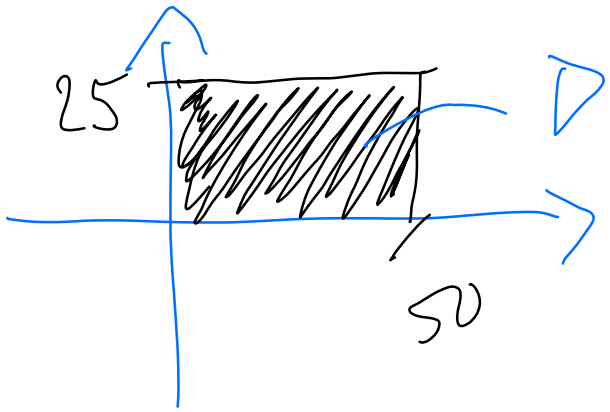
$= 567 - 207 - 126$

$= 360 - 126$

$= 234$

$$\begin{array}{r} 27 \\ \times\ 21 \\ \hline 27 \\ 540 \\ \hline 567 \end{array} \qquad \begin{array}{r} {\scriptstyle 2}\,{\scriptstyle 1} \\ 69 \\ \times\ 3 \\ \hline 207 \end{array}$$

Step 3: check boundary of region



D is a rectangle with boundary $x=0$, $x=50$, $y=0$, $y=25$

$x=0$: $f = 96y - 9y^2$.

$0 = f' = 96 - 18y \Rightarrow f'(6) = 0$,

so f inc on $[0,6]$ & dec on $y \geq 6$, so maximum is at $y=6$,

and $f(0,6) = 6(96 - 9\cdot 6) = 6\cdot 42 =$

252

$y=0$: $f = 48x - x^2$, max at

$0 = f' = 48 - 2x \Rightarrow x = 24$,

$f|_{x=24} = (48-24)\cdot 24 = 576$

$x=50$: $f = 50(48-50) - 100y + 96y - 9y^2 = -9y^2 - 4y - 100 <$ 0 for any $y \geq 0$, so not making any profit here

$y=25$: $f = 48x - x^2 + 25(96 - 9.25) - 50x = -x^2 - 2x - \text{[illegible]} < 0$ for any $x \geq 0$, so not making any profit here.

So the largest profit is \$576,000,

which happens at $x=24$, $y=0$.

2a: max change of $f=\sin(xy)$ at $(1,0)$ & direction it occurs

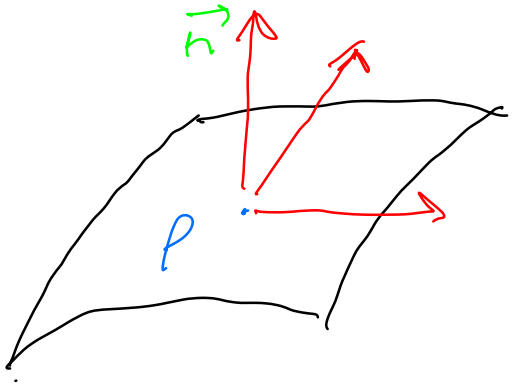
Recall: max rate of change happens in direction of ∇f at the point

$\nabla f = \langle y\cos(xy), x\cos(xy)\rangle$,

so $(\nabla f)(1,0) = \langle 0, 1\rangle$, so

max rate of change happens towards $\langle 0,1\rangle$ & it is $\nabla f \cdot \nabla f = \langle 0,1\rangle \cdot \langle 0,1\rangle = \boxed{1}$.

3: Tangent plane & parametric equations for normal line to surface of $xy^2z^3=8$ at $P=(2,2,1)$



Recall: for a surface $c = F(x,y,z)$ for some F, the vector $\nabla F(x_0,y_0,z_0)$ will be normal to the graph of the surface at (x_0,y_0,z_0).

Step 1: apply ∇F formula to get normal vector

Notice surface is $F=8$ for $F(x,y,z) = xy^2z^3$, and

$$\nabla F = \langle y^2z^3, 2xyz^3, 3xy^2z^2 \rangle,$$

so $(\nabla F)(2,2,1) = \langle 4, 8, 24 \rangle$ is a normal vector to the plane.

Step 2: convert vector to line

So $\langle 4(t-2), 8(t-2), 24(t-1) \rangle$ is a vector equation for the normal line, so

$$\left\{ \begin{array}{l} x = 4(t-2) \\ y = 8(t-2) \\ z = 24(t-1) \end{array} : t \in \mathbb{R} \right\}$$ is

normal line parametric equations

Step 3: get tangent plane from point on plane & normal vector.

$(2,2,1)$ is on the plane & $\langle 4, 8, 24 \rangle$ is a normal vector to the plane since normals to surface & plane are the same, so

$$4(x-2) + 8(y-2) + 24(z-1) = 0$$

is tangent plane equation, or

$$(x-2) + 2(y-2) + 6(z-1) = 0$$

1c: $h = x^2y^3 - 4xz$ DD at $(1,1,1)$ in direction of $v = \langle -1, 2, 0 \rangle$

Recall: DD at point p in direction of unit vector $\vec{v}$ is $(\nabla h)(p) \cdot \vec{v}$

So first, we need to normalize v.

$$\hat{v} = \frac{v}{\|v\|} = \frac{1}{\sqrt{1^2+2^2+0^2}}\vec{v} = \frac{1}{\sqrt{5}}\langle -1, 2, 0 \rangle.$$

Now we compute ∇h at point

$$\nabla h = \langle 2xy^3 - 4z, 3y^2z^2, -4x \rangle,$$

so $(\nabla h)(1,1,1) = \langle -2, 3, -4 \rangle$,

so DD is $(\nabla h)(1,1,1) \cdot \hat{v} =$

$$\langle -2, 3, -4 \rangle \cdot \frac{1}{\sqrt{5}} \langle -1, 2, 0 \rangle =$$

$$\frac{1}{\sqrt{5}}(2+6+0) = \frac{8}{\sqrt{5}} = \boxed{\frac{8\sqrt{5}}{5}}$$

46: critical pts of $g = (y-2)x^2 - y^2$ & use 2nd derivate test to classify

Step 1: Solve $0 = \nabla g$ to find points

$$\langle 0, 0\rangle = \nabla g = \langle 2x(y-2),\ x^2 - 2y\rangle$$

$0 = 2x(y-2) \Rightarrow x = 0$ or $y = 2$

$x = 0$: $2y = x^2 = 0 \Rightarrow y = 0$

$y = 2$: $x^2 = 2y = 4 \Rightarrow x = \pm 2$,

so $(0, 0)$, $(\pm 2, 2)$ are critical.

$$H = \begin{bmatrix} 2(y-2) & 2x \\ 2x & -2 \end{bmatrix}, \text{ so}$$

Hessian $= f_{xx}f_{yy} - (f_{xy})^2 =$

$$-4(y-2) - (2x)^2 = -4[(y-2) + x^2]$$

So now we have to substitute

$$Hs(0,0) = -4[-2+0] = 8 > 0,$$

so $(0,0)$ is local minimum

$$Hs(\pm 2, 2) = -4[0 + (\pm 2)^2] = -16 < 0$$

so $(\pm 2, 2)$ are saddle points.